

Biochemistry of dietary, functional and structural carbohydrates



BY

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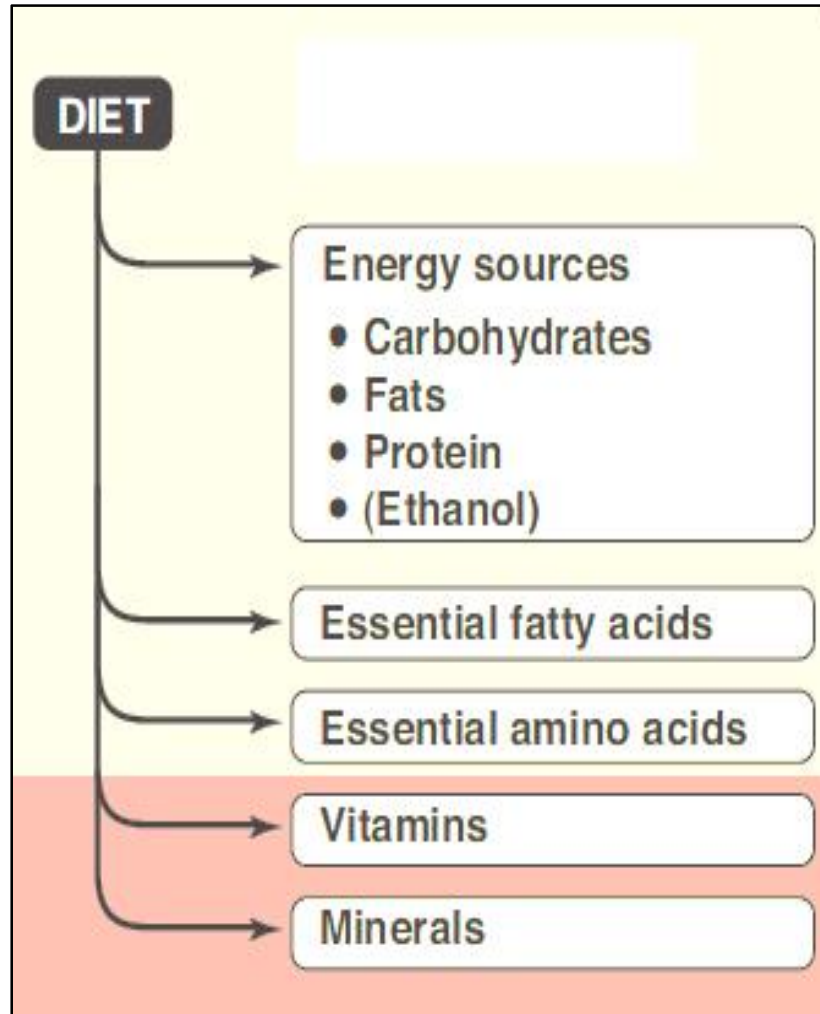
By the end of this lecture, the student will be able to



- A1. List importance of carbohydrates
- A2. Point out classification of carbohydrates
- A3 List classification of monosaccharides
- A4. Point out disaccharides and their importance
- A5. List classification of polysaccharides and their function

Nutrients and their requirements

There are about 40 dietary components, which are essential for human metabolism. Nutrients are classified as



Nutrients

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graph TD; A[Nutrients] --> B[Macronutrients]; A --> C[Micronutrients]; B --> D[compounds that are needed in amounts of a gram or more per day (which include e.g. carbohydrates, lipid and proteins).]; C --> E[compounds or elements needed in smaller quantities on the order of a milligram or less per day (which include e.g. vitamins, minerals and trace elements).];
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Macronutrients

compounds that are needed in amounts of a gram or more per day (which include e.g. carbohydrates, lipid and proteins).

Micronutrients

compounds or elements needed in smaller quantities on the order of a milligram or less per day (which include e.g. vitamins, minerals and trace elements).

1- Macro nutrients:

Include compounds that are needed in amounts of a gram or more per day e.g. carbohydrate, protein and fat.

Function of macronutrients

- 1- They are used as source of energy**
- 2- They are used as sources for essential amino acids and fatty acids.**
- 3- They are used as structural components of cells.**

2- Micro nutrients:

Include compounds or elements needed in smaller quantities on the order of a milligram or less per day e.g. vitamins, minerals

Micronutrients generally function as coenzymes or cofactors in metabolic reactions.

Sources of energy:

The dietary sources of energy include **carbohydrates**, **lipids** and **proteins**. Proteins and carbohydrates yield 4 cal/gram, lipid yield 9 cal/gram



Carbohydrates are the primary and preferred source of energy for the body.

Major food sources include breads, cereals, fruits and sugars.

At least **60 grams** of carbohydrate are needed daily by adults to provide exogenous glucose substrate for carbohydrate-dependent tissues, such as the **brain** and **erythrocytes**.

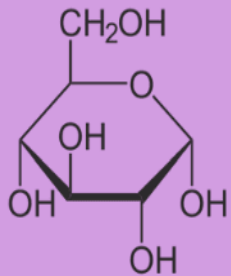
Importance of carbohydrates:

- 1- Glucose is the most important dietary carbohydrate and the major metabolic fuel to the body.
- 2- Glycogen is a storage form of carbohydrates in liver and muscles
- 3- Ribose and deoxyribose are important for synthesis of nucleic acids (RNA&DNA)
- 4- Galactose is important for synthesis of lactose of milk.
- 5- Fructose is the fuel sugar for sperms
- 6- Oligo- and polysaccharides are important constituents of cell membranes
- 7- Mucopolysaccharides are important components of connective tissues and heparin is an anticoagulant

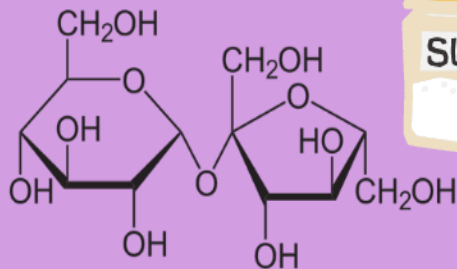
Examples of Carbohydrates

Carbohydrates are organic molecules containing carbon, hydrogen, and oxygen.

Sugars

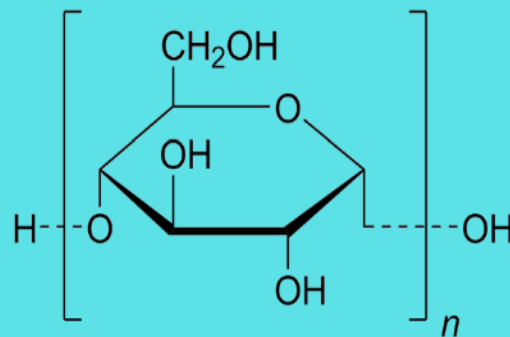


glucose



sucrose (table sugar)

Oligosaccharides



α -1,4

$2 < n < 20$

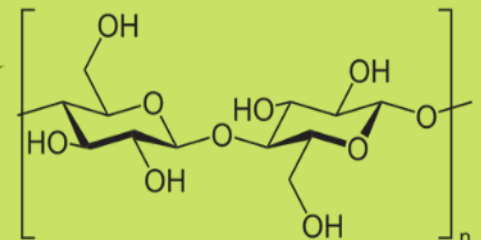


maltodextrin

Polysaccharides



glycogen



cellulose

Classification of carbohydrates

- **Monosaccharides:** They consist of a single sugar unit and cannot be further hydrolyzed.
- **Disaccharides:** They are composed of 2 sugar units.
- **Oligosaccharides:** They are formed of (3-10) monosaccharides linked by glycosidic linkage and give monosaccharides on hydrolysis
- **Polysaccharides:** They are formed of more than 10 monosaccharides linked by glycosidic linkage and give monosaccharides on hydrolysis.

Carbohydrates: Definition, Classification, Functions

Monosaccharides

Simple sugar

Aldoses	Ketoses
Glyceraldehyde	Dihydroxyacetone
Erythrose	Erythrulose
Ribose	Ribulose
Glucose	Fructose
Glucoheptose	Sedoheptulose

Disaccharides

2 monosaccharide units

Reducing

- Maltose
- Lactose

Non-Reducing

- Sucrose

Oligosaccharides

3-10 monosaccharide units

- Raffinose
- Stachyose
- Verbascose

Polysaccharides

>10 monosaccharide units

Homopolysaccharides	Heteropolysaccharides
Starch	Agar, Agarose
Dextrin	Gum & Pectin
Glycogen	Hyaluronic acid
Cellulose	Chondroitin sulfate
Inulin	Keratan Sulfate
Dextran	Dermatan Sulfate
Chitin	Heparan Sulfate



Classification of Monosaccharides

Their classification depends upon either:

1- The presence of aldehyde or ketone groups into:

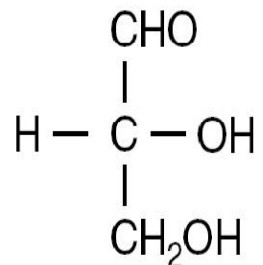
- ❑ **Aldoses** e.g : ribose & glucose
- ❑ **ketoses** e.g.: ribulose & fructose.

2- The number of carbon atoms as:

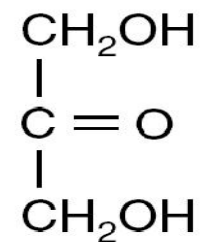
(1) Trioses:

These monosaccharides contain three carbon atoms. Trioses are formed in the course of the breakdown of glucose by glycolysis .

Trioses



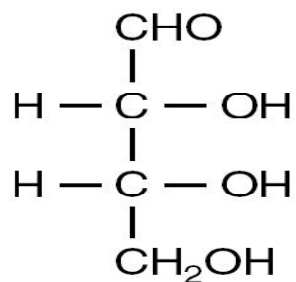
Glyceraldehyde



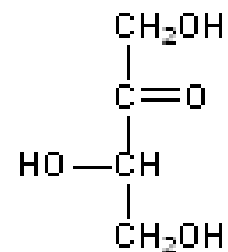
Dihydroxyacetone

(2) Tetroses:

These contain four carbon atoms



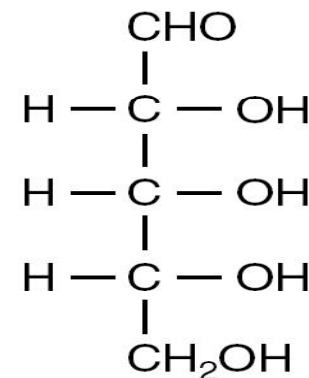
Erythrose



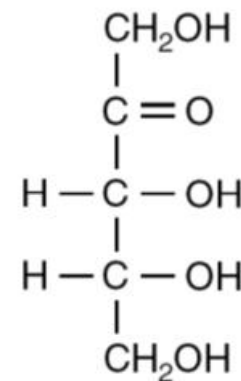
Erythrulose

(3) Pentoses

- They are monosaccharides that contain 5 carbon atoms.
- **A- Aldopentoses: D-Ribose**
- **B- Ketopentose:D-Ribulose**
- **Importance**
 - Ribose is the most important pentose because it enters in the structure of **DNA** (deoxyribose) and **RNA** (ribose) and important free nucleotides such as **ATP**, **GTP** and other high energy phosphate compounds and coenzymes **NAD** , **NADP** and **FAD**, **FMN**.



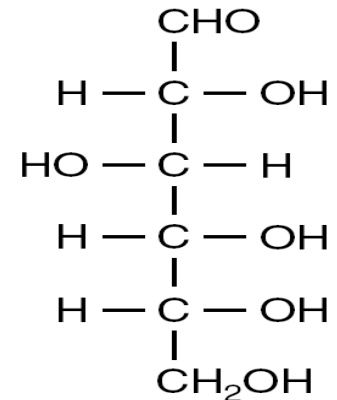
Ribose



Ribulose

(4) Hexoses

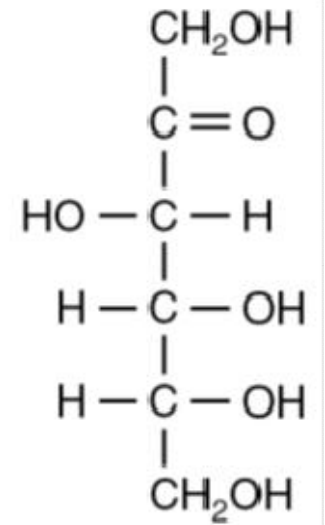
- ✓ Are monosaccharides containing 6 carbon atoms. They are the most important monosaccharides particularly glucose.
- ✓ All monosaccharides have to be converted into **glucose** to be utilized in the body.
- ✓ In cases of diabetes mellitus, glucose is raised in the blood (hyperglycemia) and appears in urine (glucosuria)



D-Glucose

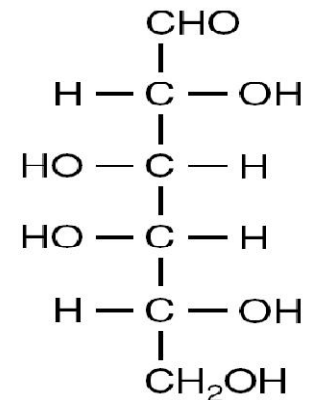
Fructose:

- ✓ It is present in fruit juices and honey.
- ✓ Fructose can be changed to glucose in the liver.
- ✓ It is the main sugar present in seminal fluid as a source of energy to sperms



Galactose:

- ✓ Galactose can be metabolized to glucose.
- ✓ It is synthesized in the mammary gland to make lactose of milk.



Classification of Monosaccharides

No. of Carbon	Type of sugar	Aldoses	Ketoses
3	TRIOSES	Glyceraldehydes	Dihydroxyacetone
4	TETROSES	Erythrose	Erythrulose
5	PENTOSES	Ribose, Xylose	Ribulose, xylulose
6	HEXOSES	Glucose, Galactose	Fructose

Sugars exhibit various forms of isomerism.

Isomers: are compounds having different structures but have the same chemical formula e.g. glucose, galactose, mannose and fructose are isomers having the same chemical formula $C_6H_{12}O_6$

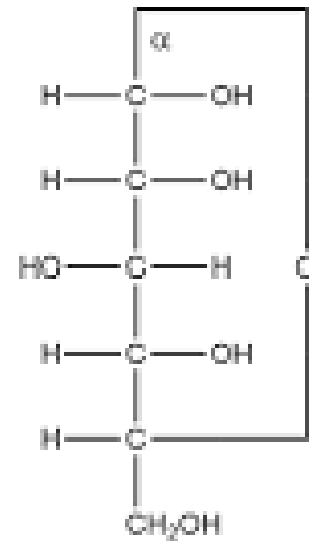
(1) Alpha and beta anomers:

These are isomers differing in the configuration of -OH and -H on C1.

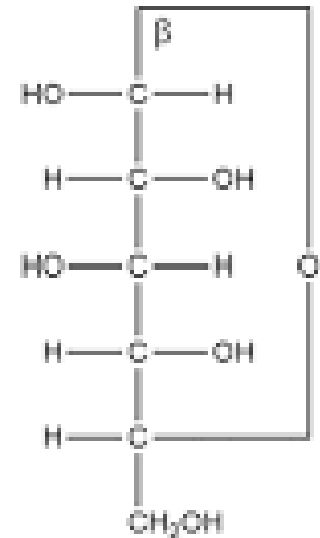
N.B:

OH on right side (α - sugar)

OH on left side (β - sugar).



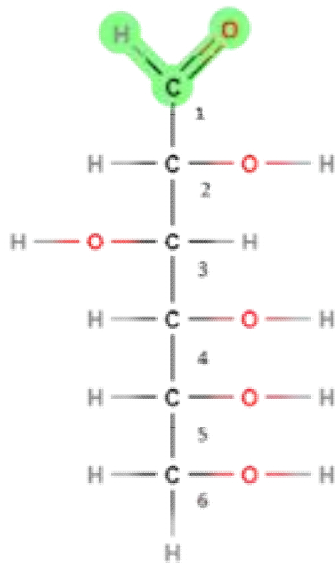
α - (D) - (+) - Glucose



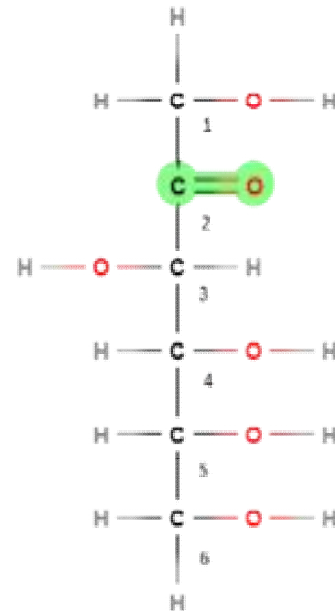
β - D - (+) - Glucose

(2) Aldose-ketose isomerism

Fructose has the same molecular formula as glucose, but fructose has ketone group in C2, whereas there is an aldehyde group in C1 of glucose



Aldose



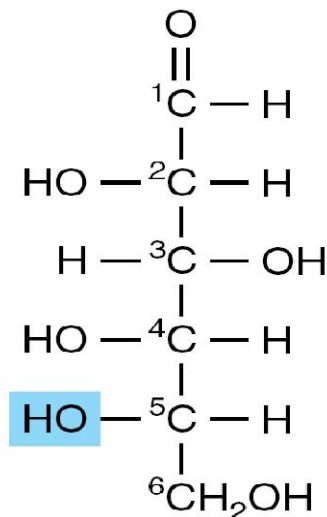
Ketose

3) D and L isomers:

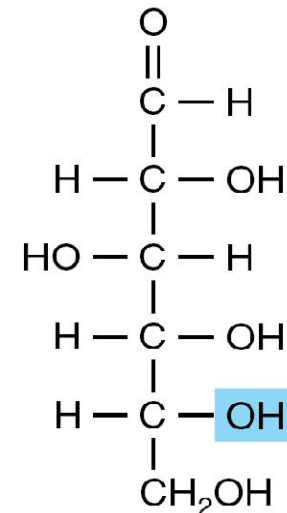
This type of isomerism depends upon the orientation of the -H and -OH groups around the carbon atom adjacent to the terminal primary alcohol carbon (**C5**).

When the -OH group is on the right, the sugar is a member of the D-glucose, when it is on the left, It is a member of the L-glucose.

Most of the monosaccharides occurring in mammals are of the D-form.



L-Glucose



D-Glucose

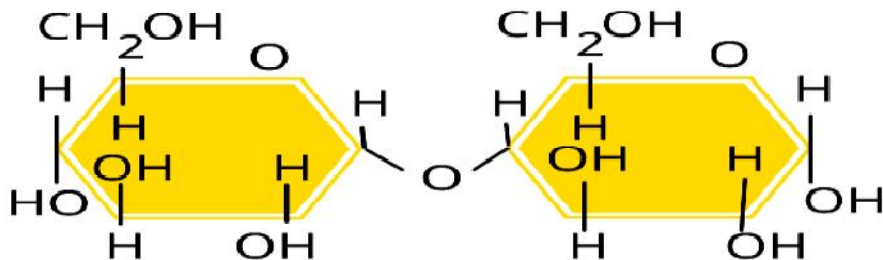
II- Disaccharides

- ❖ These are sugars formed of 2 monosaccharide units linked by a glycosidic linkage.

(1) Maltose:

- ❖ It is composed of 2 alpha glucose units united by alpha 1-4 glycosidic linkage.
- ❖ Maltose is produced during hydrolysis of starch by **amylase** enzyme.
- ❖ It is digested by **maltase** enzyme into 2 glucose.

Disaccharide Examples



Maltose
glucose + glucose

(2) Sucrose

- ❑ It is called cane sugar or table sugar.
- ❑ Sucrose is formed from a molecule of alpha- glucose with a molecule of beta- fructose united by α 1-2 glycosidic linkage.
- ❑ It is hydrolyzed by **sucrase** enzyme to glucose and fructose

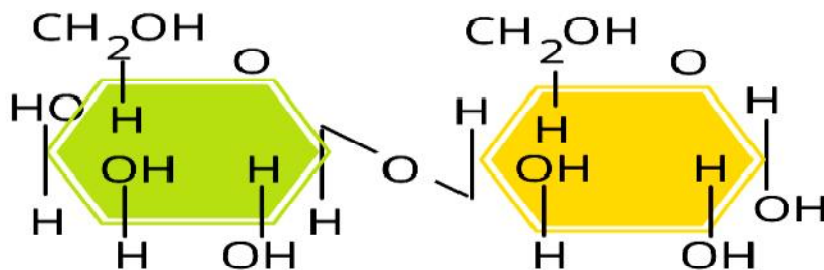


Sucrose
glucose + fructose

(3) Lactose (milk sugar):

- ✓ It is formed of a molecule of beta- galactose with a molecule of beta-glucose united by beta-1-4-galactosidic linkage.
- ✓ It is found in milk and may occur in urine during pregnancy (Lactosuria).
- ✓ Lactose is hydrolyzed by the enzyme **lactase**.
- ✓ In lactase deficiency, malabsorption of lactose leads to diarrhea and flatulence.

Disaccharide Examples



Lactose
galactose + glucose

Polysaccharides

- Polysaccharides are formed by condensation of more than 10 monosaccharide units.
- They are classified into 2 groups:
 - (A) Simple or homopolysaccharides.
 - (B) Mixed or heteropolysaccharides.

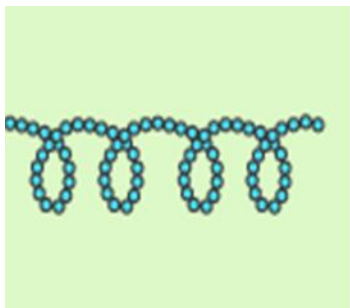
(A) Homopolysaccharides:

These are formed of only one type of sugar units. Hexosans are formed of hexoses e.g. glucosans.

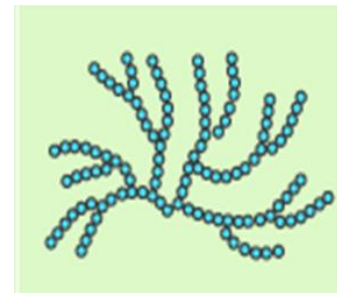
Homopolysaccharides include the following physiologically important carbohydrates:

(1) Starch(Plant sugar)

- ✓ Starch is the most important food source of carbohydrate and is found in cereals, potatoes and other vegetables.
- ✓ It is a homopolymer of α -glucose (glucosan) forming an α -glucosidic chain.
- ✓ It consists of amylose (15–20%), which is non-branching helical structure and amylopectin (80–85%), which consists of branched chains of glucose residues united by α -1 $\rightarrow$ 4 linkages in the chains and by α -1 $\rightarrow$ 6 linkages at the branch points.



Amylose

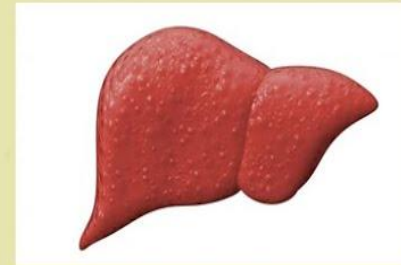
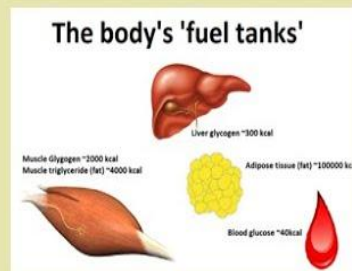


Amylopectin

(2) Glycogen (animal sugar):

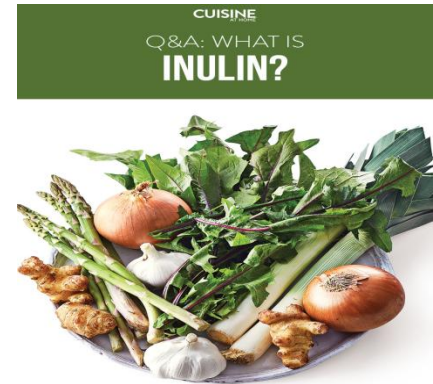
- ✓ is a homopolymer of α -glucose (**glucosan**) forming an α -glucosidic chain.
- ✓ It is the storage polysaccharide in animals (liver and muscles)
- ✓ Its structure is the same as starch but it is **highly branched**

GLYCOGEN



(3) Inulin (Fructosan)

- ✓ Inulin is fructosan as it is hydrolyzable to fructose.
- ✓ It is used for determination of glomerular filtration rate as it is completely excreted by the renal tubules.



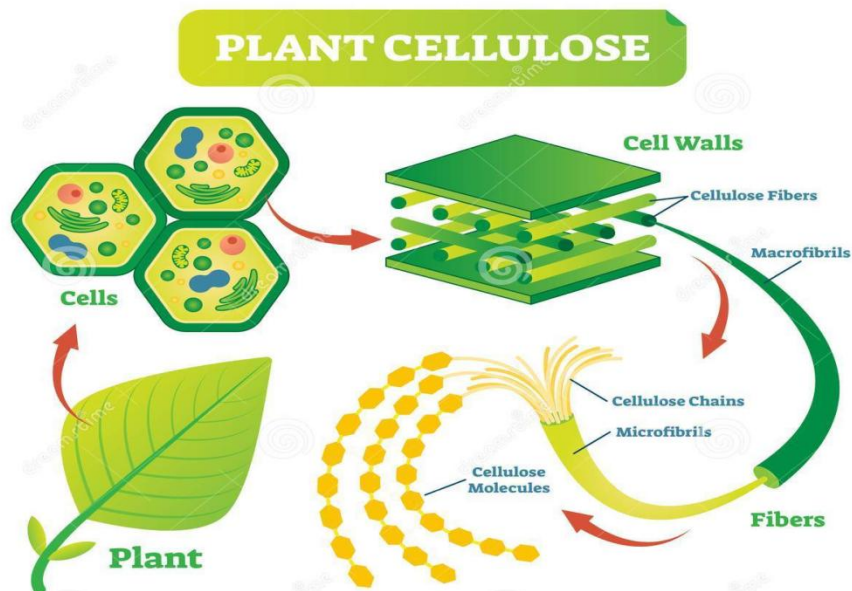
(4) Dextrins

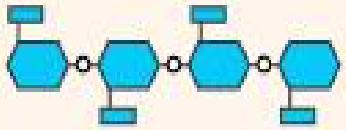
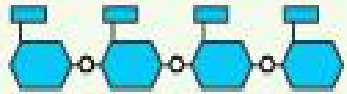
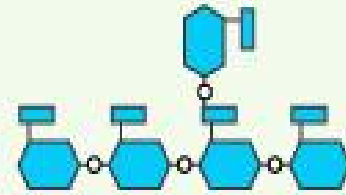
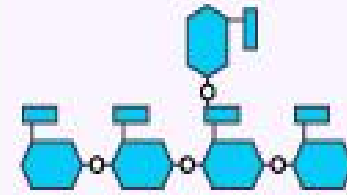
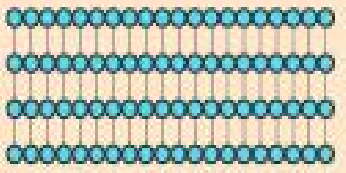
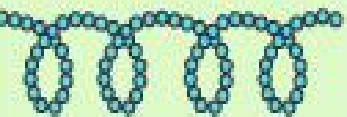
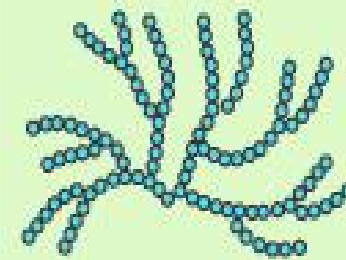
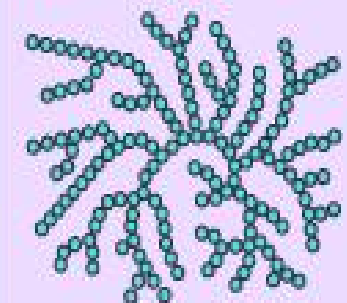
- These are substances formed in the course of hydrolysis of starch.
- At first, limit dextrins are formed. Next, erythrodextrin is formed. Finally, achrodextrins are formed which give no color with iodine.



(5) Cellulose:

- ✓ Cellulose is the chief constituent of the cell wall of plants. It is insoluble and consists of beta- glucose units linked by **beta (1-4)** glycosidic bonds.
- ✓ Cellulose can not be digested by humans because of the absence of cellulase enzyme (β -amylase) that attacks the beta-linkage, It increases the bulk of stool which increases the intestinal movement, preventing constipation.



	Cellulose	Starch		Glycogen
		Amylose	Amylopectin	
Source	Plant	Plant	Plant	Animal
Subunit	β -glucose	α -glucose	α -glucose	α -glucose
Bonds	1-4	1-4	1-4 and 1-6	1-4 and 1-6
Branches	No	No	Yes (~per 20 subunits)	Yes (~per 10 subunits)
Diagram				
Shape				

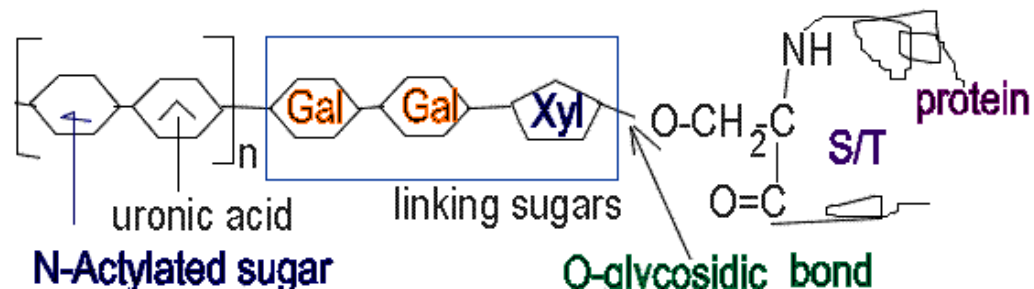
(B) Heteropolysaccharides:

These are polysaccharides formed of more than one type of simple sugar units. These include: **glycosaminoglycans** and **glycoproteins**.

(A) Glycosaminoglycans (Mucopolysaccharides):

- Glycosaminoglycans consist of chains of repeating disaccharides units which are characterized by their content of amino sugars (glucosamine or galactosamine) and uronic acids (glucuronic acid or iduronic acid).
- When these chains are linked to a protein molecule, the compound is known as **proteoglycan**.
- Glycosaminoglycans are associated with bone, elastin and collagen which are the structural elements of the tissues.
- Examples include the following:

GAG LINKAGE TO PROTEIN IN PROTEOGLYCANS



(1) Hyaluronic acid:

- ✓ Hyaluronic acid is widely distributed among various animals and tissues including synovial fluid, the vitreous humour of the eye and loose connective tissues.
- ✓ The hyaluronic acid in connective tissue can be hydrolyzed by an enzyme called **hyaluronidase** or spreading factor. This enzyme is secreted by certain bacteria and is present in the acrosomal cap of the sperms helping the process of fertilization.

(2) Heparin:

- ✓ Heparin is found in the granules of mast cells and also in liver, lung and skin.
- ✓ It has anticoagulant properties
- ✓ It activates plasma lipoprotein lipase enzyme for hydrolysis of triglycerides.

(3) Chondroitin sulfates:

- ❖ It is present in cartilage, tendons, ligaments, skin, aorta and cornea.
- ❖ Function: in cartilage, it binds collagen and hold fibers in strong network, help to maintain the shape of skeletal muscles and have a role in compressibility of cartilage in weight bearing.

(B) Glycoproteins:

Glycoproteins are proteins containing carbohydrates in varying amounts (2-15 units which is usually branched)

The constituent carbohydrates include:

- ✓ **Hexoses- Pentoses- Methyl pentoses- Sialic acids:** (N-acetylneuraminic acid).

Functions of Glycoproteins:

- Structural function e.g. cell walls, collagen, elastin, fibrin and bone matrix.
- Lubricant and protective functions e.g. mucins and mucous secretions.
- Immunologic function e.g. immunoglobulin.
- Hormonal function e.g. thyroid stimulating hormone (TSH).
- Catalytic function e.g. proteases and nucleases
- Cell attachment and recognition sites e.g. cell-cell, virus-cell, bacterium-cell and hormone receptors.

Proteoglycans

- Mainly carbohydrates 85% with 15 percent protein.
- Composed of REPEATING units – heteropolysaccharide
- Carbohydrates are linear with more than 50 sugar units

Glycoproteins

- Mainly protein with minimal carbohydrates
- Composed of varying monosaccharide units.
- Carbohydrates are branched with 3-10 sugar units.

1- Which one of the following sugars is found in DNA?

a- Deoxy-ribose

b- ribose

c- ribulose

d- xylulose

2- Which one of the following sugars is a disaccharide ?

a- Starch

b-Lactose

c- Galactose

d- Glucose

3- Carbohydrates that are formed of (3-10) sugar units linked by glycosidic linkage are called:

a- Monosaccharides

b- Oligosaccharides

c- Disaccharides

d- Polysaccharides

4- Which one of the following is considered ketose?

a- Glucose

b- Glyceraldehyde

c- Fructose

d- Erythrose

5- Dietary carbohydrates yieldenergy/gram

- a. 4 cal
- b. 7 cal
- c. 9 cal
- d. 12 cal

Reference

- 1- Harpers illustrated Biochemistry
- 2- Lippincott illustrated reviews integrated systems

Lecture 3 Biochemistry of CHO.mp4

